

**APPENDIX H — EXAMPLE EMBODIMENT COMBINATIONS AND VARIANTS
(NON-LIMITING)**

A. Overview

This appendix provides non-limiting example combinations of nodes, workflows, and processing variants. Any embodiment described herein may be implemented with fewer nodes, additional nodes, substituted sensor modalities, or altered workflow sequencing while maintaining the functional intent of privacy-preserving outputs, confidence labeling, cross-device correlation, and cross-user correlation under consent gating.

B. Embodiment 1 — Two-Node Longitudinal + Context System (Non-Limiting)

In one example embodiment, the system comprises: 1. a longitudinal intimate-region node configured to collect time-series data and generate privacy-preserving metrics; and 2. an upper-body context node configured to generate context state labels and confidence indicators. The computing system aligns streams, gates outputs to valid windows, and presents trend curves with confidence indicators.

C. Embodiment 2 — Baseline-First Architecture (Non-Limiting)

In one example embodiment, the system uses one or more session-based baseline nodes prior to longitudinal tracking. A geometry node and/or a compliance node generates baseline references used to normalize longitudinal indices. The computing system then performs longitudinal tracking using a wearable node, producing stability and change-over-time metrics referenced to the baseline.

D. Embodiment 3 — Longitudinal-First With Later Baseline (Non-Limiting)

In one example embodiment, the system begins with longitudinal tracking to collect repeated windows and establish an initial within-user reference distribution. A baseline acquisition session is performed later using a geometry and/or compliance node. The computing system reconciles baseline outputs with historical longitudinal metrics by applying normalization transforms and versioning the baseline reference.

E. Embodiment 4 — Geometry + Anti-Spoof Enhanced Scan (Non-Limiting)

In one example embodiment, a session-based external geometry node performs scan-path validation using IMU constraints and sensor agreement. The system computes scan confidence and anti-spoof confidence using physiologic confirmation and contact integrity checks. Outputs comprise privacy-preserving profiles, zone summaries, and repeatability scoring.

**F. Embodiment 5 — Intraluminal Geometry/Compliance With Stimulus-Response Curves
(Non-Limiting)**

In one example embodiment, an intraluminal node applies controlled radial stimulus and measures response via pressure transducers and circumferential contact sensing. The computing system computes compliance indices, contact stability indices, hysteresis proxies, and validity labels. Outputs are stored as baseline references and used for later correlation workflows.

G. Embodiment 6 — Multi-Node Correlation Suite (Non-Limiting)

In one example embodiment, the system combines three or more nodes and computes cross-device correlation across derived metrics. Correlation outputs include alignment scores, synchrony curves, lag signatures, context-matched scores, stability indicators, and combined confidence indicators.

H. Embodiment 7 — Cross-User Correlation With Derived-Metric Exchange (Non-Limiting)

In one example embodiment, cross-user correlation is performed under consent gating with derived-metric exchange. Each user's device computes privacy-preserving representations locally, and the computing system computes alignment and synchrony scores without sharing raw signals. Outputs include consent status indicators, share-scope indicators, and confidence indicators based on quality thresholds.

I. Embodiment 8 — Local-Only Mode (Non-Limiting)

In one example embodiment, all processing is performed on-node and/or on a user device, and no remote storage is required. Outputs remain privacy-preserving and include confidence indicators. Cross-user correlation is disabled or limited to local representations unless user-controlled consent is enabled.

J. Variations

Any embodiment may use substituted sensor modalities, altered duty cycles, alternative processing pipelines, different windowing strategies, and different privacy modes. The embodiments may be adapted for consumer-facing or clinical-facing presentation while maintaining privacy-preserving output formats.